

PokerFace: Trading the Gap Between What CEOs Say and How They Say It

Gator Quant Hacks 2026 · Systematic Trading · quant note · code: github.com/ian-hoang/donttouchmynonosquare · pre-registration commit 6fe0912

1. Summary

We turn public CEO interviews into a daily equity signal. For 1,833 curated YouTube interviews of 20 mega-cap CEOs (2016–2026) we extract facial action proxies (MediaPipe face mesh, 52 blendshapes), voice quality (openSMILE eGeMAPS) and word use (Loughran–McDonald, Larcker–Zakolyukina categories) from the CEO's own segments only, score each against that CEO's last 30 interviews, and combine the cues with **signs and weights fixed in advance from meta-analytic effect sizes**. We short high-stress interviews and buy calm ones, beta-hedged with SPY, for 20 sessions. In-sample (2016–Sep 2024, 661 trades) the strategy's net Sharpe is **0.07** (costs $\times 2$: -0.04); on the sealed out-of-sample period (Oct 2024–Sep 2026, evaluated once) it is **-0.33**. **Neither number survives scrutiny**. In-sample the strategy fails our own pre-registered kill condition (FF5+UMD alpha $t = 0.09$); out-of-sample the signal has the wrong sign (rank IC -0.066). Two candidates that looked promising in-sample (voice-only, QQQ hedge; both ≈ 0.2) also turn negative once the sealed window is opened. Our conclusion is a well-powered null: across 1,813 interviews and 661 + 234 trades, the stress visible in mega-cap CEOs' faces and voices is already in the price at a 20-day horizon. Every follow-up we registered after the failure also fails: 192-variant optimization (winner -0.87 out-of-sample), walk-forward selection (-0.54), the same signal on 20 less-watched CEOs, and post-interview drift. Since October 2024 the score's sign has reversed in both universes (pooled IC -0.12), which we flag as the next hypothesis to test rather than trade.

2. Economic hypothesis

Who is on the other side. Attention-driven investors who trade on a CEO's scripted verbal message and on-camera persuasion. CEO appearances on CNBC lift prices about 79 bp on the day and give back about 108 bp within ten days (Kim & Meschke); persuasive delivery moves capital independently of content (Hu & Ma, JF); analysts underweight negative vocal affect on earnings calls (Mayew & Venkatachalam 2012, JF). **Why it persists.** Involuntary vocal, facial and verbal leakage of concealed bad news (Koo, Teoh, Yoo & Zhao 2026, where cross-modal incongruence in 2,439 CEO interviews predicts negative earnings surprises and restatements; Hobson, Mayew & Venkatachalam 2012; Larcker & Zakolyukina 2012) is *soft* information: costly to process at scale (Engelberg 2008), arriving under limited attention (DellaVigna & Pollet 2009; Hirshleifer, Lim & Teoh 2009), and noisy per event (human lie detection is $\sim 54\%$, Bond & DePaulo 2006), so no single trade is an arbitrage. The news then comes out gradually through guidance and the next earnings report. **Hypothesis (written before results).** *Mega-cap stocks whose CEO publishes an interview with a high tell score (stress and verbal–nonverbal incongruence relative to the CEO's own past) underperform a beta-hedged benchmark over the next 20 sessions, and low-tell interviews do the reverse.* We measure stress against a person's own baseline; **we do not claim to detect lies**. It fails if the in-sample top-minus-bottom spread is ≥ 0 , alpha $t < 1$, the effect is explained by the pre-interview run-up, a placebo built from folk cues (gaze aversion, blinking, fidgeting) works as well, or the out-of-sample sign flips.

What is new. Koo et al. (2026) link CEO-interview incongruence to accounting outcomes using a commercial affect API. We build the measurement from open-source models, normalize against each CEO's own history, weight cues by published effect sizes fixed in advance, and test a **tradeable, cost-aware, beta-hedged strategy with a sealed out-of-sample period**.

3. Data and universe

Universe. 20 CEOs of US large caps chosen by role and video availability, never by returns, inside their verified CEO tenures and after listing (Cook ends 2026-08-31, Iger 2026-03-17, Gelsinger 2024-12-01, Saylor 2022-08-08; all date corrections are logged in docs/DEVIATIONS.md). **Corpus.** Two rounds of YouTube search (4,938 + 10,418 hits; round 2 added year-specific queries because round 1 skewed to 2023–26) → 5,531 after rule filters → 2,082 kept by LLM curation agents that saw **metadata only, never prices** (unscripted Q&A only; keynotes, compilations, re-uploads, AI fakes and "about the CEO" videos removed) → 1,833 dated inside tenure windows → 1,813 processed → 1,596 with the CEO's face verified. **Event time** is the YouTube upload timestamp (UTC, to the second); a re-upload can only make an event later. **Prices:** Yahoo Finance daily OHLCV, split- and dividend-adjusted (we trade open-to-open; a returns fingerprint is committed). **Factors:** Ken French FF5 + momentum. **Earnings dates:** SEC EDGAR 8-K Item 2.02 filings. Interviews within ± 2 sessions of an earnings filing are excluded (142 events). **Survivorship:** Intel (Gelsinger) and MicroStrategy (Saylor) are kept for their tenures; the universe is today's well-known CEOs, which biases toward names that did well, so every trade is hedged and the signal is long–short *within* each CEO's own history (name-level drift cancels).

4. Methodology

Measurement (per video, 12-minute window). On UF HiPerGator: MediaPipe face mesh (52 blendshapes, head pose) and pose at 4 fps; Whisper large-v3-turbo on GPUs. ArcFace identifies the CEO as **the face that recurs across that CEO's interviews**; an audio-visual vote over ≤ 15 s utterances assigns speech to the CEO. **Cues** (signs, weights $\propto$ meta-analytic $|d|$; config/signal_spec.json): face: chin raise (+0.25), lip press (+0.16), facial pleasantness (−0.12), illustrator gestures (−0.085); voice: pitch (+0.195), jitter/shimmer/HNR as vocal-tension proxies; text: negations, third-person plural, general-knowledge references, extreme positive emotion (+), anxiety, numbers, shareholder value (−); plus a verbal–nonverbal incongruence term (upbeat words $\times$ tense face). Cues that do *not* replicate in meta-analyses (gaze aversion, blinking, self-touch, head motion, smiling, pauses, fillers, speech rate) are excluded and form the FOLK placebo. **Normalization.** Each cue is a robust z-score against the same CEO's ≤ 30 strictly earlier interviews (≥ 5 required). TELL is the equal-weight mean of the face, voice, text and incongruence scores (≥ 5 cues across ≥ 2 modalities). **Signal and trade.** $S = -\text{TELL} / \sigma(\text{TELL of all earlier events})$. Decision time is upload + 2 h; we enter at the next regular-session open and exit 20 sessions later (open-to-open); a new interview replaces that CEO's open position. Size: a full-signal position targets 10% annualized idiosyncratic volatility ($\leq 35\%$ notional per event, $\leq 50\%$ per name, $\leq 200\%$ gross), hedged with SPY at the 252-day ex-ante beta. **Costs:** 5 bps/side on stocks, 1 bp on SPY, 30 bps/yr borrow; costs $\times 2$ always reported. **Inference:** Newey–West FF5+UMD alpha, bootstrap CIs, Deflated Sharpe over 23 logged trials, permutation and random-date nulls. **No lookahead, enforced:** a test suite rewrites all data after t and asserts that no signal, weight or return before t changes; run_all.py refuses to run if it fails.

5. Results

net of costs	in-sample (2016-01 – 2024-09)	out-of-sample (2024-10 – 2026-09)
events traded	661	234
annualized return	0.3%	-3.7%
volatility	9.5%	10.0%
Sharpe (gross / net / costs $\times 2$)	0.20 / 0.07 / -0.04	-0.17 / -0.33 / -0.47
max drawdown	-33.3%	-14.2%
turnover ($\times$ capital / yr)	34.13	44.81
FF5+UMD alpha (NW t)	0.3% (0.09)	-1.1% (-0.18)
rank IC, S vs 20-day CAR	0.025	-0.066

In-sample. The signal points the right way but is weak: high-S (calm) interviews beat low-S (stressed) ones by 0.31% over 20 days (CEO-clustered 95% CI includes zero), the rank IC is 0.025, and the hit rate is 49.9%. Every neighboring horizon (1–60 sessions) has a negative Sharpe, so 20 days is an isolated value rather than a plateau. Kill condition triggered: $\alpha t < 1$. **Out-of-sample** (opened once, per config/oos_plan.json committed beforehand): net Sharpe -0.33, rank IC -0.066. The spread has the wrong sign, which is our second kill condition. The two post-hoc candidates fell from in-sample Sharpes of 0.21 (voice-only) and 0.22 (QQQ hedge) to -0.23 and -0.15, a clean example of in-sample selection that does not generalize. **What the data do show.** By modality, voice was the only positive block in-sample and text was strongly negative (Sharpe -0.56): the Larcker–Zakolyukina "deceptive" word categories predict *higher* subsequent returns in TV interviews, the opposite of their sign on earnings calls. The unconditional post-interview drift is also positive in our sample (long every interview: in-sample Sharpe 0.28), unlike the 10-day reversal Kim & Meschke report for CNBC; we did not pre-register it, so it is reported as an open question and was not tested out-of-sample.

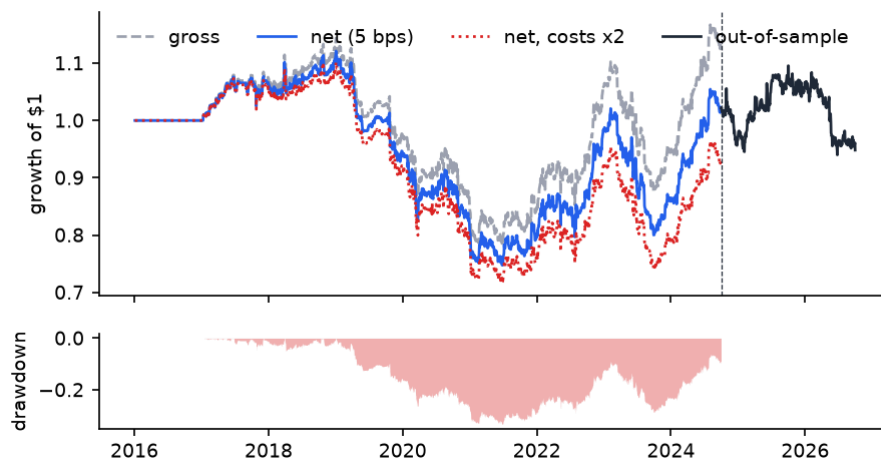


Figure 1. Growth of \$1: in-sample gross, net and costs $\times 2$, then the sealed out-of-sample period; drawdown below.

Optimization does not rescue it. We searched 192 in-sample variants (8 horizons $\times$ 2 hedges $\times$ 6 signal blocks $\times$ both signs). The best in-sample Sharpe is 0.32 (text_only|-|h20|SPY), below the 0.66 that the best of 192 *useless* strategies would reach by luck; the CSCV probability of backtest overfitting is 85.7%, and the in-sample winner earns -0.87 on the sealed window. Walk-forward optimization (each year trade the best variant of all prior years) earns a Sharpe of -0.54 over 2018–2026; last year's winner lost money in 8 of 9 years. On these names there is no stable version of the signal to find. **Tests against chance.** Deflated Sharpe 0.15 over 23 logged trials; PSR(SR>0) 0.59;

bootstrap 95% CI for the in-sample Sharpe [-0.57, 0.72]. Shuffling S within each CEO beats the actual Sharpe with probability 0.06; moving every event to a random day of the same year, 0.17. **Pre-registered placebo:** the FOLK-cue score earns a Sharpe of -0.11 (IC -0.011). **Run-up control:** with the 5-day pre-entry run-up in the regression, S has $t = 0.62$ (corr with run-up -0.08). **Mechanism (prediction 6):** high TELL and subsequent idiosyncratic volatility, CEO-clustered $t = 0.48$ ($n = 667$).

Follow-up tests, pre-registered after the primary failed (HYPOTHESIS_2.md, committed before their data were collected or opened). **H2, less-watched CEOs:** the identical frozen signal on 20 new, lower-attention CEOs (946 interviews, 298 + 142 trades): net Sharpe 0.14 (2016–2024-09) and -0.92 (2024-10 onward), rank IC -0.004 / -0.212, FF5+UMD alpha $t = 0.21$. **H2 fails:** the in-sample-period IC is zero and in 2024-10 onward the spread has the *opposite* sign and is significant (calm minus stressed -7.91%, CEO-clustered 95% CI excludes zero). Pooled over all 40 CEOs the rank IC is +0.017 before October 2024 and -0.120 after, in both universes. **Our reading:** after 2024 the score behaves like *arousal* (excited CEOs whose stocks keep running), not concealment. Every holdout is now used, so we do not trade the inverted sign; it is the first hypothesis we would pre-register on 2026-10 onward data. **H3, post-interview drift versus the same stocks on random days:** excess 20-day CAR -0.27% (95% CI [-0.014962481815354006, 0.01046492575378207]) on the original universe's sealed window and 0.18% on the H2 universe; the in-sample "buy every interview" result was survivorship, not an interview effect.

6. Risk management

Limits (pre-set, in config/strategy.json). Each event is sized to 10% annualized idiosyncratic volatility at full signal, capped at 35% notional; any one name at 50%; gross at 200% including the hedge. A new interview replaces the CEO's existing position, so exposure never stacks on one name. **Market and factor risk.** Every leg is hedged with SPY at its 252-day ex-ante beta. On FF5+UMD the strategy's market beta is 0.00, momentum beta -0.02, and R^2 0.01: returns are not a disguised factor. **Tail and regime.** Daily 99% VaR 1.79%, CVaR 2.46%, worst day -3.86%; 49.5% of months positive; Calmar 0.01. Stress windows (COVID crash, 2022 bear market, SVB, Aug-2024 vol spike) are in the appendix. **De-risking rule.** A drawdown brake (halve exposure while drawdown exceeds 10%) is pre-declared and reported as a variant; it is off in the headline so the headline is the simplest rule. **Concentration.** Results by CEO and leave-one-CEO-out runs (appendix) check that no single CEO drives the result.

7. Liquidity and capacity

The universe is the most liquid US equities (median ADV of names traded $\approx$ \$1,285M). At \$10M AUM a typical trade is 0.03% of ADV. We charge 5 bps per side; with a one-cent tick these names quote half-spreads of roughly 0.1–0.3 bp, so 5 bps per side also covers opening-auction slippage and impact at this size (SEC fee 0.2 bp; general-collateral borrow $\approx$ 25 bps/yr, D'Avolio 2002; we charge 30). Turnover is $34.13\times$ capital per year. With square-root impact ($\text{cost} = \text{fixed} + 0.7\cdot\sigma\cdot\sqrt{(Q/ADV)}$), the in-sample gross edge is so small (gross Sharpe 0.20) that costs absorb half of it at about \$8M of AUM. **The binding constraint is the signal, not liquidity:** at \$10M a typical trade is 0.03% of ADV, so an edge of the size the literature suggests would scale to hundreds of millions in these names.

8. Why it failed, limitations, and next steps

Candidate explanations. (1) *Efficiency where attention is highest:* these are the 20 most-watched CEOs in the world, and every interview is dissected by analysts and retail investors, which is exactly

where underreaction to soft information should be smallest. (2) *Wrong outcome or horizon*: the closest prior work (Koo et al. 2026) finds that interview incongruence predicts accounting outcomes (negative earnings surprises, restatements) rather than 20-day returns; the price effect may arrive only at the next earnings release, mixed with everything else. (3) *Timing*: YouTube upload time can trail broadcast by hours, so some reaction happens before our entry. (4) *Measurement noise*: 12-minute windows at 360p and automated identity and speaker attribution. **Limitations.** Survivorship in the universe (mitigated by hedging and within-CEO normalization); 14 videos lost their audio. **Next steps.** Test the same pipeline on (a) accounting outcomes and the next earnings surprise, (b) earnings-call video for the Russell 1000, where attention is lower, (c) broadcast airtimes with minute bars, and (d) the unconditional post-interview drift as its own pre-registered hypothesis.

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- Data: YouTube; Yahoo Finance (yfinance); Kenneth R. French Data Library; SEC EDGAR. Tools: MediaPipe (Google), openSMILE (audEERING), faster-whisper, InsightFace ArcFace, UF HiPerGator.

Appendix (outside the 5-page limit)

A1. Robustness (all pre-declared, in-sample, each logged as a trial).

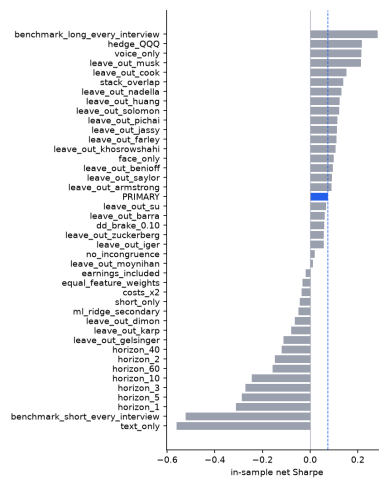


Figure A1. In-sample net Sharpe of every pre-declared variant; the primary specification in blue.

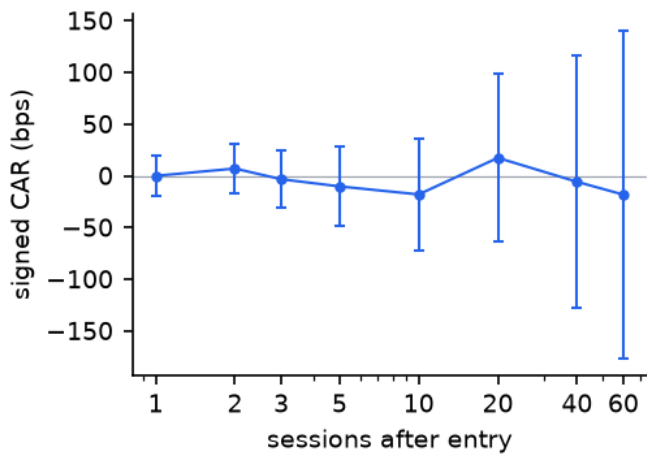


Figure A2. Signal decay: mean signed beta-hedged CAR by horizon (event level, 95% CI).

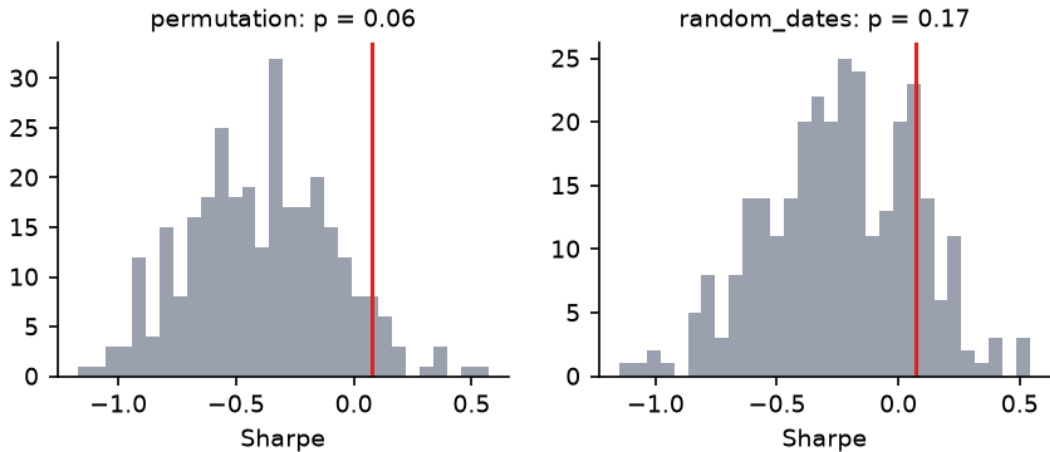


Figure A3. Null distributions: Sharpe under within-CEO permutation and random event dates; red line = actual.

A2. Stress windows (strategy vs. SPY, in-sample).

window	dates	strategy	SPY	strategy max DD
2018Q4 selloff	2018-10-01 to 2018-12-24	2.0%	-18.7%	-3.1%
COVID crash	2020-02-19 to 2020-03-23	-8.4%	-30.2%	-9.6%

window	dates	strategy	SPY	strategy max DD
2022 bear market	2022-01-03 to 2022-10-12	8.4%	-25.8%	-8.7%
SVB crisis	2023-03-08 to 2023-03-24	-4.6%	0.3%	-4.9%
Aug-2024 vol spike	2024-07-16 to 2024-08-07	5.8%	-6.9%	-1.2%

A3. Pre-committed case studies. Tell score per interview against the stock's price relative to SPY. *Illustration, chosen with hindsight, not evidence:* of four ledger cases with interviews in the 180 days before a later-contradicting event, only one stands out. Pat Gelsinger's most stressed interview in our sample (100th percentile of his own history) came on 2024-04-08, three days after he told CNBC Intel expected its foundry business to break even in 2027; INTC fell about 19% beta-hedged over the next 20 sessions, and on 2024-08-01 Intel suspended its dividend. Cook (before the January 2019 revenue warning) and Musk show nothing unusual.

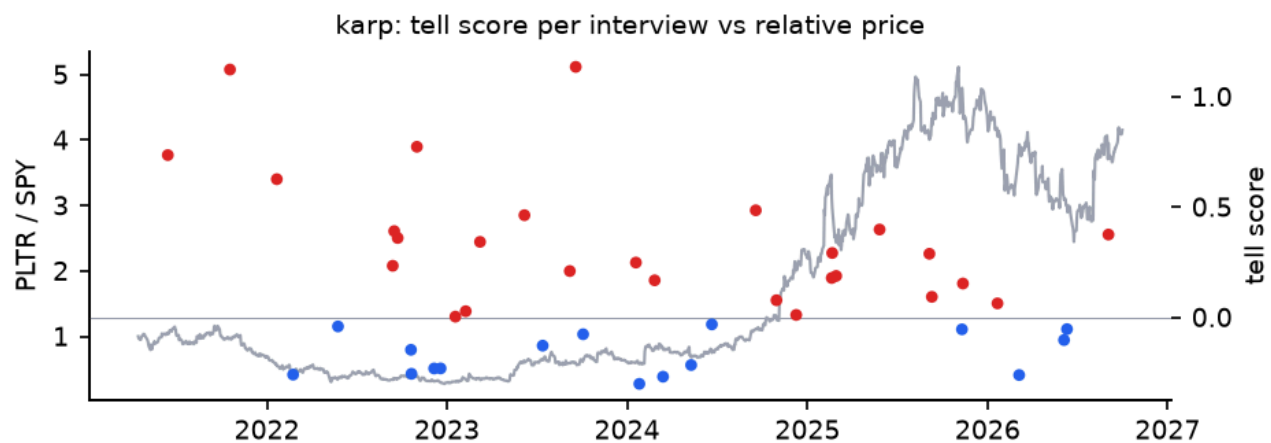


Figure A4. Alex Karp (PLTR).



Figure A5. Elon Musk (TSLA).

A4. Deviations from the pre-registration. Every post-registration change, with time and reason, is listed in docs/DEVIATIONS.md: corpus round 2 (year-specific search, same curation rules), verified CEO tenure corrections (Cook, Iger), and pipeline fixes made before any in-sample result. None was informed by returns. **A5. Exploratory (not pre-registered): what CEOs say × how they say it.** LLM agents labeled masked CEO answers as claims, denials, promises and hedges (data/features/utterance_labels.csv). Despite masking, agents could identify the speaker in 76.8% of videos, so these labels could carry hindsight and are kept out of the trading signal. 20-day

CAR after interviews with a denial: 0.86% (n = 370); denial + high tell: 1.08% (n = 121); denial + low tell: -0.01% (n = 117).